

A Zero-Parameter Latent Color Operator: Photometric Equivariance for Diffusion Tokenizers

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Abstract

We derive a *zero-parameter, closed-form* latent-space operator for photometric transforms brightness, contrast, saturation, hue on diffusion tokenizers. These transforms, unlike the spatial warps of prior equivariance work, have no canonical action on the latent grid. We canonicalize this augmentation family as a subgroup of the pointwise color-affine group $\text{Aff}(3)$ and obtain the operator by conjugating the pixel-space action through the fitted linear latent-to-RGB map of SD-VAE. On the frozen autoencoder this operator steers brightness, contrast, and saturation roughly $2\times$ past the do-nothing floor but fails entirely on hue, even though a per-image decoder-inversion oracle shows every factor is decoder-expressible (ΔE_{00} ceilings 1.7–5.5). Twenty thousand steps of decoder-routed equivariance fine-tuning close this gap: the closed-form operator reaches ΔE_{00} 4.4–4.9 on *all four* factors closing 0.76–0.92 of the oracle gap against a compute-matched control at a cost of 0.31 dB PSNR. Learned operators (a Lie-algebra parameterization and a strictly more expressive convolutional variant) provide no consistent overall advantage, and an intervention that matches the latent spectrum without the equivariance constraint confers none of the equivariance. Diffusion transformers trained on the equivariant latents converge faster at two model scales (27% lower FID at the 40K checkpoint for DiT-S; 14–18% at 30–40K for DiT-B, replicated across two seeds) with equal final quality.

1 Introduction

Latent diffusion models [7, 8] generate in the latent space of a pretrained autoencoder, and a growing body of work shows that the *structure* of that latent space not only its reconstruction quality governs how fast and how well diffusion trains. EQ-VAE [3] demonstrated that fine-tuning the autoencoder so that *spatial* transforms of the image (scaling, 90° rotation) correspond to the same transforms applied directly to the latent grid yields large diffusion speedups. Parallel work regularizes for scale [10] and shift [13] equivariance. In every one of these cases the latent operator is *inherited*: a spatial warp of an image has an obvious counterpart on the spatially downsampled latent.

Photometric transforms break this pattern. A brightness change, a saturation shift, or a hue rotation acts pointwise on RGB values, and a diffusion tokenizer’s latent channels are not RGB: there is no canonical answer to the question “*what does a hue shift do to a latent?*” This paper treats that missing operator as the central object of study.

Contributions.

1. **A closed-form latent color operator.** We canonicalize the standard photometric family (brightness, fixed-anchor contrast, saturation, hue) as a subgroup/monoid of $\text{Aff}(3)$, prove an exact photometric–spatial commutation lemma, and derive a zero-parameter latent operator

$M_a = W^+ A_a W + (I - W^+ W) K$ from the fitted linear latent-to-RGB map W , with an explicit treatment of W ’s null space (Section 3).

2. **An oracle methodology.** Per-image decoder inversion gives, for each transform, the best equivariance error *any* latent edit could achieve. This turns the headline result into a fraction of the achievable rather than a raw error, and separates two failure modes that are otherwise confounded: on the frozen SD-VAE, hue is decoder-expressible ($\Delta E_{00} \approx 2.9$ oracle) yet unreachable by affine steering (at the do-nothing floor).
3. **The closed form is sufficient once the autoencoder cooperates.** Brief decoder-routed fine-tuning makes the closed-form operator approach the oracle ceiling on all four factors, including hue. A learned Lie-algebra operator, its ablations, and a convolutional operator with 90K parameters provide no consistent overall advantage over it. Single-factor training reveals a structured transfer pattern; and a spectral-matched intervention shows that, at the representation level, the equivariance is not a side effect of latent spectral statistics.
4. **Faster diffusion convergence at two scales.** DiT-S and DiT-B trained on the equivariant latents reach mid-training FID milestones substantially earlier than compute-matched controls, echoing at the photometric level the convergence benefits previously reported for spatial equivariance, with endpoint quality unchanged (two seeds per arm at the larger scale).

2 Related Work

Equivariance regularization of diffusion latents. EQ-VAE [3] fine-tunes pretrained autoencoders so that scaling and 90° rotations of the input correspond to grid operations on the latent, routing the constraint through the decoder after observing that explicit latent matching collapses the representation; we inherit both the loss structure and that constraint. Skorokhodov et al. [10] regularize for scale equivariance via joint downsampling; Zhou et al. [13] for shift equivariance. DC-AE 1.5 [1] structures latents by channel-wise masking and explicitly forgoes equivariance; VA-VAE [11] aligns latents with vision foundation models. All prior equivariance work in this line is spatial; the operator-design problem that photometric transforms pose is, to our knowledge, unaddressed.

Learned group actions and color equivariance. The Homomorphism AutoEncoder [2] learns a group representation acting on the latent of a small autoencoder trained from scratch on group-structured data; our learned baseline transplants this idea onto a production-scale tokenizer (and finds it unnecessary). CEConv [5] builds hue equivariance into network architecture for discriminative tasks; we instead regularize a pretrained tokenizer, requiring no architectural change.

Skepticism about equivariance as a latent property. Zhong et al. [12] report, across 86 heterogeneous tokenizers, that an affine-equivariance metric correlates weakly with diffusion quality. Our setting complements theirs: we hold architecture, data, and compute fixed, vary only the constraint, and include a spectral-matched intervention, an in-family interventional probe rather than a cross-family correlation.

3 Method

3.1 A canonicalized photometric family

We work with pixel colors $x_p \in \mathbb{R}^3$ and define the photometric family as pointwise color-affine maps $\tau_a(x)_p = A_a x_p + b_a$ with $a = (A_a, b_a) \in \text{Aff}(3)$. The standard augmentation factors embed as one-parameter families: *brightness* $A = \beta I$; *contrast* $A = \gamma I$, $b = (1 - \gamma)g\mathbf{1}$ anchored at fixed gray $g = 0.5$; *saturation* $A = sI + (1 - s)\mathbf{1}w^\top$ with BT.601 luma weights w ; *hue* a rotation $R(\theta)$ about the luma axis (YIQ convention). Two canonicalizations are load-bearing. First, library implementations of contrast use the *image-dependent* mean gray, which is not a pointwise map and breaks composition; the fixed anchor restores an exact group law. Second, saturation is a *monoid* with a non-invertible absorbing element at $s=0$ (grayscale). Equivariance targets are computed before clipping to $[0, 1]$; we measure the clipped-pixel fraction rather than assuming it small (it reaches 20–27% at the extremes of the widest ranges, which is why training uses $\beta, \gamma \in [0.7, 1.3]$).

For any pointwise-affine latent map T and any spatial warp S implemented with convex-combination interpolation, $T \circ S = S \circ T$ exactly, since affine maps commute with convex combinations. Photometric latent operators therefore compose freely with the spatial operators of prior work, without ordering ambiguity.

3.2 The closed-form operator

SD-VAE latents $z \in \mathbb{R}^{4 \times 32 \times 32}$ admit a well-known approximate linear color decoding $\hat{x}_p \approx Wz_p + c$; we re-fit (W, c) by least squares on 1,024 OpenImages validation images and obtain $R^2 = 0.859$ (per channel 0.844/0.890/0.847). If decoding were exactly this pointwise map, equivariance $D(T_a z) = \tau_a(D(z))$ would force, per latent site, $WT_a(z_p) + c = A_a(Wz_p + c) + b_a$. Solving for a channel-affine $T_a(z_p) = M_a z_p + m_a$ with W of full row rank yields

$$M_a = W^+ A_a W + (I - W^+ W) K, \quad m_a = W^+ (A_a c + b_a - c), \quad (1)$$

where W^+ is the Moore–Penrose pseudo-inverse and K acts on the one-dimensional subspace invisible to the fitted linear RGB probe. We use $K = I$ (identity on the residual subspace), for which the homomorphism $M_{a \circ b} = M_a M_b$ holds exactly on the color block. The operator has *no trainable parameters*: for a sampled transform, compose the factor matrices in homogeneous coordinates and apply Eq. (1).

3.3 Decoder-routed equivariance training

Following Kouzelis et al. [3], the constraint reaches the latent only through the decoder: with probability $p=0.5$ per sample we replace the reconstruction objective by $\mathcal{L}_{\text{rec}}(\tau_a(x), D(T_a(E(x))))$, never including any latent-matching term (the documented collapse mode). The posterior is pushed forward through the affine operator analytically (mean $M_a \mu + m_a$; marginal standard deviation $\sqrt{\text{diag}(M_a \text{diag}(\sigma^2) M_a^\top)}$), and the KL term is computed on the untransformed posterior. The fine-tune is GAN-free ($\ell_1 + \text{LPIPS} + \text{KL}$), 20K steps at batch 8, AdamW 10^{-4} .

3.4 The oracle: decoder-inversion ceilings

For each factor and magnitude we optimize a latent z' directly to render the transformed target, $\min_{z'} \|D(z') - \tau_a(x)\|^2$ (500 Adam steps, 16 images). The resulting error is the decoder’s expressivity ceiling which is an empirical upper bound from unconstrained per-image latent optimization and the R^2 of a channel-affine fit from $E(x)$ to z' measures how affine the oracle’s edit is. Oracle edits are not unique, so this R^2 is a lower bound on affine representability.

Table 1: Equivariance error (mean ΔE_{00} , lower is better) and reconstruction after 20K fine-tuning steps. All rows except B0 are compute-matched. Stored- W operator evaluation; the fair B1 reference for gap-closed uses a re-fitted W reference 15.5/9.6/14.5/16.3 (see text), not the stored- W B1 row. Bold: best per column. Seed repeats put the per-factor spread at ± 0.3 – $0.6 \Delta E_{00}$, so smaller differences are not meaningful; among the photometric conditions, brightness is the only factor on which the closed form separates from the learned variants beyond that noise.

Condition	Bright.	Contr.	Sat.	Hue	Composed	PSNR	rFID
B0 (frozen)	7.24	5.71	7.04	10.21	5.15	25.35	13.7
B1 (control)	31.6	11.6	23.6	24.6	17.4	25.34	17.3
B2LITE (spatial)	29.5	10.6	21.7	23.0	15.6	24.71	19.0
C1PROXY (spectral)	35.9	16.5	28.9	33.7	22.2	25.12	18.0
P1 (closed form)	4.59	4.58	4.40	4.92	4.10	25.03	18.4
P2 (learned Lie)	6.50	5.01	5.70	4.95	4.47	25.08	18.0
P2–comp. loss	6.44	4.81	5.08	4.52	4.04	25.11	17.6
P2–analytic init	6.26	4.49	4.34	4.82	3.93	25.20	18.1
P3 (conv, 90K params)	10.0	4.75	3.81	4.65	3.93	25.22	17.8
Oracle ceiling (frozen)	2.2–5.5	2.1–4.3	1.7–3.0	2.8–2.9	—	—	—

4 Experiments

4.1 Setup

All experiments use SD-VAE (`sd-vae-ft-mse`, $f=8$, 4 channels) and the OpenImages validation split (41,620 images at 256^2 ; 4). Equivariance error (EE) is the decoder-side discrepancy $\Delta E_{00}(D(T_a E(x)), \tau_a(x))$ in CIEDE2000 [9], averaged over a factor-specific grid including held-out magnitudes and held-out compositions; CIEDE2000 is used for color-fidelity thresholds because LPIPS is comparatively insensitive to smooth global color shifts. Reconstruction is reported as PSNR and rFID at $n=1,024$ (comparable across rows only). Every fine-tuned condition, including the control B1, receives identical steps and losses; the frozen autoencoder B0 is a reference row, never the comparison baseline. Seed repeats of B1 and P1 agree within 0.3 – $0.6 \Delta E_{00}$ per factor.

4.2 Tier 0: the frozen autoencoder

On the frozen SD-VAE the closed-form operator beats the do-nothing floor by roughly $2\times$ for brightness (7.9 vs. 17.0 ΔE_{00} at $\beta=0.6$), contrast (5.4 vs. 9.2), and saturation (7.3 vs. 12.6 at $s=0$), but sits *at* the floor for hue (13.0 vs. 14.4 at -45°). The oracle explains why this is an operator problem rather than a decoder problem: inversion reaches ΔE_{00} 2.2–5.5 across factors and magnitudes (hue: 2.8–2.9 at $\pm 45^\circ$) so every factor is decoder-expressible and the oracle’s latent edits are 79–90% channel-affine, with magnitude rather than factor driving the non-affinity.

4.3 Fine-tuning closes the gap with a zero-parameter operator

Table 1 is the main result. After fine-tuning, the closed-form operator reaches ΔE_{00} 4.4–4.9 on all four factors hue drops from worse-than-nothing to 4.92, closing 82% of the gap to its 2.9 frozen-decoder oracle ceiling at a reconstruction cost of 0.31 dB PSNR against the control. Because plain fine-tuning itself drifts the latent color geometry (the control’s EE under a W re-fitted on its own latents is 15.5/9.6/14.5/16.3), we compute the headline *gap-closed* fraction $(EE_{B1} - EE_{P1}) / (EE_{B1} - EE_{\text{oracle}})$

against that re-fitted reference, with all three terms averaged over the magnitudes at which the oracle was run (the range endpoints, where P1’s EE is 4.8/4.7/4.6/5.3): 0.92/0.76/0.82/0.82 per factor.

Three additional observations. (i) The spatial condition B2LITE validates the training loop against the known effect direction (its *spatial* EE is 6.1 vs. the control’s 10.5) while leaving photometric EE essentially unchanged as spatial equivariance does not incidentally confer photometric equivariance. (ii) All fine-tunes cost rFID relative to the frozen B0 (13.7 $\rightarrow$ 17–19): the GAN-free recipe’s price, shared equally by the control. (iii) Photometric training mildly degrades spatial EE (13.5 vs. 10.5), so the two equivariances trade off rather than come for free together.

What does not help. None of the added mechanisms we tested provide consistent overall advantage over the closed form. The learned Lie operator matches the closed form on hue and is slightly worse elsewhere; removing its composition loss or analytic initialization changes nothing beyond seed noise; and the 90K-parameter convolutional operator improves saturation slightly while degrading brightness (10.0 vs. 4.6) so no evidence the latent color action is spatially heterogeneous.

Single-factor transfer. Training on one factor reaches near-ceiling on that factor (dedicated hue: 4.40, the best hue of any run) and transfers partially to the others (2–4 $\times$ better than the control everywhere) without reaching their ceilings; brightness training transfers near-ceiling to contrast (both are diagonal scalings), while nothing transfers fully to hue (the chroma rotation). The transfer pattern respects the algebra of Section 3.1. Notably, single-factor runs pay no measurable reconstruction cost.

4.4 Mechanism: an intervention, not a correlation

A natural deflationary hypothesis, prompted by Zhong et al. [12], is that equivariance training merely smooths latent statistics. C1PROXY tests this by intervention: it fine-tunes with a spectral regularizer matched to P1’s measured latent spectrum (achieved: slope -0.821 vs. -0.832 ; high-frequency fraction 0.0248 vs. 0.0247) and *no* equivariance constraint. Its photometric EE is worse than the control’s on every factor under the fair re-fitted- W protocol (18.2/10.5/22.9/20.6 vs. 15.5/9.6/14.5/16.3; Table 1). Matching the spectrum confers none of the equivariance at the representation level, the effect is not spectrally mediated. A complementary diagnostic tells the same story: the linear latent-to-RGB fit quality R^2 falls from 0.86 (frozen) to 0.65 under plain fine-tuning, is *preserved* at 0.79 by equivariance training, and is destroyed hardest (0.55) by the spectral-only regularizer.

4.5 Generation: convergence at two scales

We train unconditional DiT-S/2 (33M) and DiT-B/2 (130M) [7] for 100K steps on cached latents of B1 and P1, identical hyperparameters, FID-5K [6] every 10K steps against a fixed reference of 10K identically processed OpenImages images (Figure 1). The reference is drawn from the same 41K-image pool the DiTs train on—no disjoint split was reserved—so absolute FID here measures fit to the training distribution; every claim in this section is a paired comparison in which both arms share the identical reference. At DiT-S scale the equivariant latents lead throughout the 30–50K window 147 vs. 202 at 40K (27%) and 112 vs. 121 at 50K with a larger 44% gap at 30K that partly reflects an early-FID transient in the control arm, and the arms meet at the endpoint (77.6 vs. 77.9). At DiT-B scale we ran two seeds per arm. The mid-training acceleration replicates in both (18% and 18% lower FID at 30K; 10% and 18% at 40K; pooled: 250 vs. 303 at 30K, 119 vs. 138 at 40K),

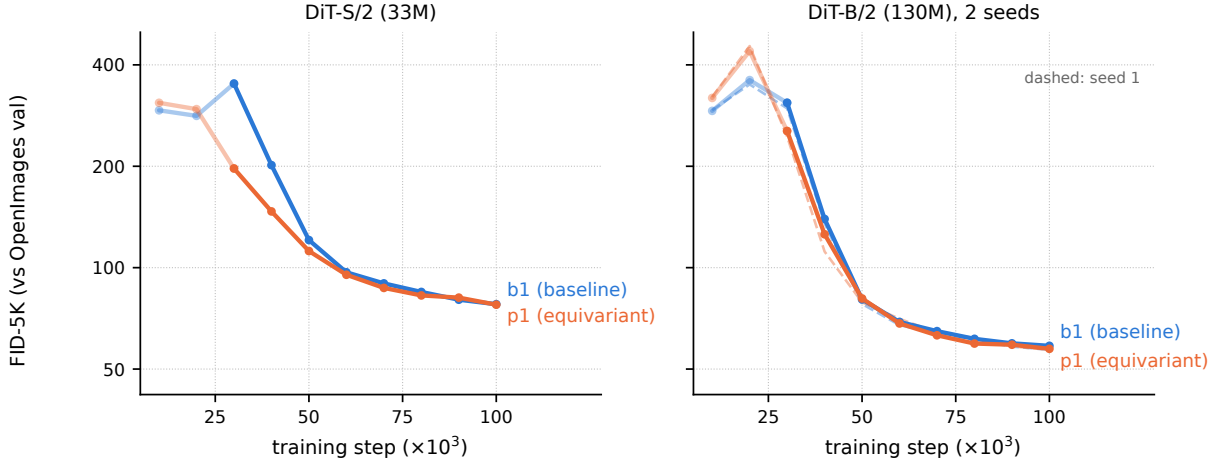


Figure 1: FID-5K versus training step for diffusion transformers trained on control (B1) versus photometric-equivariant (P1) latents, at two model scales. Unconditional generation on OpenImages latents, cfg-free DDPM sampling; points below 30K steps are shown at reduced opacity (early FID-5K is noisy in both arms). DiT-B shows two seeds per arm (dashed: seed 1); DiT-S one seed.

making the acceleration consistent at both model scales and across both DiT-B seeds. Final quality is equal: pooled endpoints 57.2 vs. 57.8, with the between-seed spread of a single arm (≈ 1.5 FID) exceeding the pooled difference; a small late-training edge visible in the first seed did not replicate in the second, and we do not claim it. The overall shape being acceleration with converging endpoints and matches what Kouzelis et al. [3] report for spatial equivariance, now at the photometric level.

5 Limitations

Our generation results are unconditional, at 41K-image scale, with FID-5K and two seeds per arm at DiT-B (one at DiT-S); class-conditional ImageNet experiments with a disjoint FID reference are the natural next step. The fine-tuning recipe is GAN-free and costs 3–5.3 rFID against the frozen autoencoder across all conditions including the control. The fixed-anchor contrast canonicalization differs from library implementations (transfer to image-mean contrast is untested). The spectral intervention rules out spectral mediation at the representation level only; generation-level mediation would require the intervention arm to be carried through diffusion training. The oracle ceilings are computed on the frozen decoder; since fine-tuning modifies the decoder, they serve as a fixed external reference rather than each condition’s own expressivity bound. Finally, all results are on one tokenizer family (SD-VAE, 4 channels); 16-channel tokenizers, where the null space of W is 13-dimensional, are unexplored.

6 Conclusion

Photometric transforms admit a simple latent-space action on diffusion tokenizers but only after the tokenizer is asked to support it. The operator itself costs nothing: it is determined in closed form by the latent-to-RGB linear map, and brief decoder-routed fine-tuning makes it nearly saturate the decoder’s expressivity ceiling on all four color factors, including hue, where the frozen model fails completely. Added learning capacity is unnecessary, spectral statistics do not explain the effect,



Figure 2: Seed-paired samples at 100K steps (DiT-B). Left: control latents; right: photometric-equivariant latents.

and diffusion models train faster on the resulting latents at two scales. We release code, evaluation batteries, and fine-tuned checkpoints.

Reproducibility and compute. All code, the evaluation battery, training configurations, and fine-tuned checkpoints are available at https://github.com/rashzxbrown/photometric_eqvae.

Each 20K-step fine-tune takes about 3.25 GPU-hours; each 100K-step DiT-B run takes about 9.5 hours on a single H200; Tier-0 probes run on a single 24 GB GPU.

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A Qualitative latent color editing

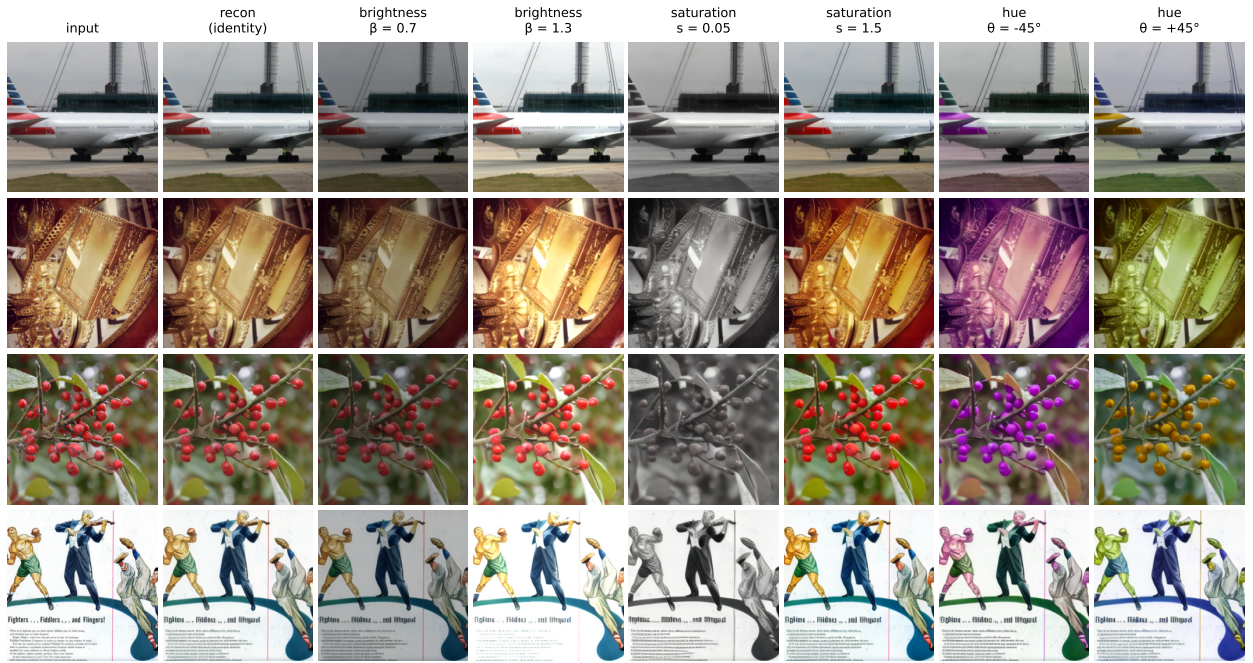


Figure 3: Latent-space color editing with the closed-form operator on the fine-tuned tokenizer: each column applies Eq. (1) for the labeled transform directly to the latent before decoding.